STAT 214 Spring 2026
Week 4

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🚨 Lab 1 Reminders 🚨

- Lab 1 is due a week from today!!! Many of you have seemingly not started 😒
- Make sure Lab 1 submission is formatted correctly
- **Do not push** /data/ folder
- Lab reports are anonymous (no name, student id, email, etc.)

Working with PSC Bridges-2

`stat-214-gsi/computing/psc-instructions.md`

Connecting to Bridges-2

- SSH Connection:
  - `ssh psc_username@bridges2.psc.edu`
- Web Interface:
  - Use Open OnDemand: **ondemand.bridges2.psc.edu**
- Reminder:
  - Login nodes are for file management, environment setup, and job submission only—do not run research code on them.

Setting Up Your Bridges-2 Environment

1. SSH to a login node:
   - `ssh psc_username@bridges2.psc.edu`
2. Load Conda
   - `module load anaconda3`
3. Create your environment:
   - `conda env create -f /ocean/projects/mth250011p/shared/214/environment.yaml`
4. Activate the environment:
   - `conda activate stat214`
5. Install the IPython kernel:
   - `python -m ipykernel install --user --name stat214 --display-name stat214`
6. Clean up Conda files:
   - `conda clean --all`

Testing Your Environment – Command Line

1. Start an interactive GPU job:
    - **`interact -gpu -t 00:10:00`**
2. Check GPU details:
    - **`nvidia-smi`**
3. Test PyTorch
    - Load Conda and activate your environment:
        - **`module load anaconda3`**
        - **`conda activate stat214`**
    - Start Python:
        - **`python`**
    - In the Python shell, run:
        - **`import torch`**
        - **`print(torch.version.cuda)`**
        - **`print(torch.cuda.is_available())`**

Testing Your Environment – JupyterLab

1. Launch Jupyter Lab via Bridges-2 Web Interface:
   a. Go to **https://ondemand.bridges2.psc.edu**
   b. Select "**Jupyter Lab: Bridges2**"
2. Set Parameters:
   a. Number of hours: 1
   b. Number of nodes: 1
   c. Account: mth250011p
   d. Partition: GPU-shared
   e. Extra Slurm Args: **`--gpus=1`**

Testing Your Environment – JupyterLab

- Verify:
  - In JupyterLab, create a notebook using the “stat214” kernel.
  - In a notebook cell, run:
    - **`import sys`**
    - **`print(sys.executable)`**
    - **`import torch`**
    - **`print(torch.version.cuda)`**
    - **`print(torch.cuda.is_available())`**
- Open a terminal in JupyterLab and run:
  - **`nvidia-smi`**

Avoiding Wasted GPU Resources

Development Workflow:

- Develop and test code locally.
- Use small datasets and shorter runs for debugging.
- Transfer code to Bridges-2 for production runs.

Resource Management:

- Use short time limits on jobs.
- Ensure jobs terminate when complete.
- Shut down idle interactive sessions.

Always shut down your Jupyter Lab server when finished (via File → Shut Down or through the web dashboard).

Only use for interactive experimentation, not for production runs.

Final Reminders

- Use the Right Partition:
  - **GPU-shared** for 1 GPU; reserve **GPU** for jobs that truly need 8 GPUs.
- Monitor Usage:
  - Check job status with **`squeue -u your_username`** and GPU usage with **`nvidia-smi`**.
- Class-Related Work Only:
  - Bridges-2 is for work directly related to this class.
- Need Help?
  - Contact a GSI if issues arise.

Judgment Call Activity: See `activity.ipynb`